B2.3 Cell specialization

NAME	DATE	TOTAL
		_____ / 32

About 30 minutes, or one section at a time. You will need a calculator. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 **STATE** two properties of stem cells. [B2.3.2] [2]

2 **STATE** two locations of stem cell niches in the adult body. [B2.3.3] [2]

3 **DEFINE** totipotent and multipotent. [B2.3.4] [2]

Section B: Understanding

12 MARKS

4 **EXPLAIN** how cells in one embryo can differentiate differently despite having the same genes. [B2.3.1] [3]

5 **OUTLINE** the role of a stem cell niche. [B2.3.3] [2]

6 **EXPLAIN** how the sizes of a red blood cell and an egg suit their roles. [B2.3.5] [3]

7 **EXPLAIN** why a low surface area-to-volume ratio limits cell size. [B2.3.6] [4]

Section C: Working with data

9 MARKS

All data in this section are invented for practice. The patterns are realistic.

8 [B2.3.6]

Agar cubes containing a pH indicator were placed in dilute acid. The time taken for each to change colour completely was recorded.

SIDE OF CUBE / MM	5	10	20
SURFACE AREA / MM^2	150	600	2400
VOLUME / MM^3	125	1000	8000
TIME TO CHANGE / S	120	410	1500

a **CALCULATE** the surface area-to-volume ratio of each cube. [3]

b **DESCRIBE** the relationship between ratio and time. [1]

c **EXPLAIN** this relationship. [2]

d **STATE** two variables that should be kept constant. [2]

e **SUGGEST** one way in which this model differs from a real cell. [1]

Section D: Longer answer

5 MARKS

9 **OUTLINE** how potency changes from a zygote to adult tissue, with examples. [B2.3.1, B2.3.4] [5]